

All you need to know about relational hyper-event modeling

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🌐: <https://github.com/martinaboschi/rem-tutorial>

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Università
della
Svizzera
italiana

Section	Presenter	Time	Abstract
Core RHEMs	Melania	13:00 – 13:45	computing hyperevent covariates; formulating, estimating, and interpreting linear RHEMs.
Practical 1	Martina	13:50 – 14:35	computing RHEM statistics using <code>eventnet</code> .
Non-Linear RHEMs	Martina	14:40 – 15:15	modeling non-linear, time-varying covariate effects while accounting for network and actor heterogeneity.
Practical 2	Martina	15:20 – 16:00	applying RHEMs to empirical datasets using R Studio.

All our materials at the following link:



<https://martinaboschi.github.io/rem-tutorial/>



NOTES

Formulary, Slides, and Notes

Supporting materials for the workshops

→



WORKSHOP 1 - PRACTICAL

Introduction to Relational Event Models

Simulating and fitting REMs

→



WORKSHOP 2 - PRACTICAL

All You Need to Know About Relational Hyper-Event Modeling

Fitting and interpreting RHEMs

→



SOFTWARE

amore package

R package

→



RESOURCE

Introduction to Relational Event Modelling

Tutorial paper

→



Not sure about the notation or need a quick review?



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During our practicals...



You can either **follow along online** or **download the materials** and run them locally!

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Practical 2: Scientific Innovation

The practical sessions of this workshop focus on **fitting relational hyper event models** using the `amor` package. If you would like to replicate the code locally, feel free to download the files in the box below:

Download files

- `Scientific-Innovation.R` — R script, for following along with the code
- `Scientific-Innovation.Rmd` — R Markdown source file
- `Scientific-Innovation.zip` — working directory with all data and image files needed to replicate the full analysis

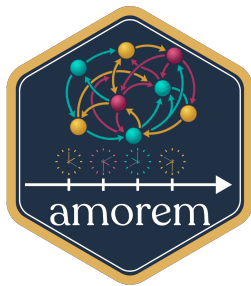
During our practicals... we will use amore package!



Newly developed R package that implements:

- ▶ simulation of relational event data
- ▶ flexible and computationally efficient inference techniques beyond linear effects

<https://francisorichter.github.io/amorem/>



Relational Event Model



Dynamic network as sequence of instantaneous time-stamped interactions:

- ▶ Relational event (s, r, t) s sends a postcard to r at time t



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How does the network unfold over time? What drives it?

- ▶ Each dyad modelled as a **counting process** $N_{sr}(t)$ = number of $s \rightarrow r$ in $[0, t]$. They have a "propensity" to occur at time t given past history $\mathcal{H}_{t-}$ - **hazard** -

$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{Is } (s, r) \text{ at risk?}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp\{f(x_{sr}(t), t)\}}_{\text{edge-specific risk factors}}$$



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- $\lambda_0(t)$: global risk drivers very few postcards sent if t is night/weekend
- $f(x_{sr}(t), t)$: edge-specific drivers, that may depend on prior interactions
 - if s and r are not close friends, then (s, r) less likely $\rightarrow$ exogenous
 - if r sent postcard to s before t , then (s, r) more likely $\rightarrow$ endogenous.

Estimating effect of drivers from data



- ▶ **Relational Data:** stream of observed events $E = \{(s_i, r_i, t_i), i = 1, \dots, n\}$
- ▶ **Parametric Model:** **linear** (in log-hazard) $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t) \dots$ just for now!
- ▶ **(Sampled Partial) Likelihood:**
 - **Data** says: "at t_i the dyad (s_i, r_i) occurred".
 - **Model** argues: "other dyads at risk ($W_{s^*r^*}(t_i) = 1$) could have occurred instead but didn't, the *non-events*".

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- Given that something happened at t_i and past history $\mathcal{H}_{t_i-}$, consider probability that it was indeed (s_i, r_i) against all other competing dyads $(s^*, r^*) \rightarrow$ too much competition, computationally intense :(

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$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i)}{\lambda_{s_i r_i}(t_i) + \lambda_{s_i^* r_i^*}(t_i)} = \prod_{i=1}^n \frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\} + \exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}$$

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AIM: estimate β that maximizes **likelihood**.

Estimating effect of drivers from data



- **(Sampled Partial) Likelihood:** **one** non-event sampled convenient because

$$\begin{aligned}
 \mathcal{L}^S(\beta) &= \prod_{i=1}^n \frac{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\} + \exp \{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} \\
 &= \prod_{i=1}^n \frac{\frac{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp \{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}}{\frac{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp \{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} + \frac{\exp \{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}{\exp \{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}} = \prod_{i=1}^n \frac{\exp \{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}}{1 + \exp \{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}} \\
 &= \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta \mathbf{x}_i) \quad \Delta \mathbf{x}_i = \mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i)
 \end{aligned}$$

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Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{\beta \cdot \mathbf{x}_{sr}(t)\}$$

Inference:

Logistic Regression

$$Y_i | \Delta \mathbf{x}_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta \cdot \Delta \mathbf{x}_i$$

$$y_i = 1 \quad i = 1, \dots, n; \quad \text{no intercept}$$

Estimating effect of drivers from data



- **(Sampled Partial) Likelihood:** **one** non-event sampled convenient because

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta x_i) \quad \Delta x_i = x_{s_i r_i}(t_i) - x_{s_i^* r_i^*}(t_i)$$

stratum	time	sender_ev	receiver_ev	sender_nv	receiver_nv	dist_ev	dist_nv
1	0.0015464	31218	31255	31419	31419	1.6821306	0.000000
2	0.0029034	31272	31324	33202	31966	1.7032925	2.873395
3	0.0096929	31937	32050	32050	31262	2.7475273	2.547647
4	0.0452324	31261	31262	31313	31218	1.1820339	2.201659
5	0.0517514	31272	31272	31419	31218	0.0000000	1.875181
6	0.0524939	33200	31261	31255	33202	0.5538851	1.243866
7	0.0531366	31285	31419	31419	33200	1.6052291	1.784903
8	0.0572758	32604	32217	31966	31262	3.0713035	2.821915
9	0.0628786	31285	33202	31071	31023	1.0922588	1.974220
10	0.0676353	31255	32050	31297	31272	2.4686079	1.655558

In R (amorem):

```
rem(~ dist, data = dat_gam, method = "gam")
```



Back to postcards being sent...

- ▶ Relationship distance $x_{sr}(t)$, aka small value $\rightsquigarrow (s, r)$ very close friends, big value $\rightsquigarrow (s, r)$ barely friends.
Estimated $\beta < 0 \implies$ the greater the distance between s and r , the less likely it is that s sends a postcard to r
- ▶ Reciprocity $x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$ as 0/1 indicator of whether r sent postcard to s before t .
Estimated $\beta > 0 \implies r$ having sent postcard to s in the past, makes s sending a postcard to r more likely in the future (in principle...)



Why is an extension needed?

We will refer to two main guiding examples:

1. **Les Misérables**: *co-appearance* of actors in one or more chapters $\Rightarrow$ each chapter is a *hyperevent*.
2. **Scientific innovation**: *publication* of an article by a group of authors citing a set of articles $\Rightarrow$ each new paper is a *hyperevent*.



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🕒 : Discussion time!

1. How do these two types of hyperevents differ?
Who are the senders, and who are the receivers?



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🕒 : Discussion time!

1. How do these two types of hyperevents differ?
Who are the senders, and who are the receivers?
2. Can you think of an example of a hyperevent with a single sender and multiple receivers?
3. How were these events treated before the introduction of RHEMs?
Do you have suggestions for converting them into dyadic relational events?

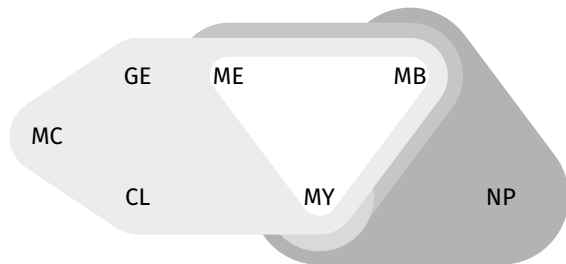


Let us look at each example: 1. *Les Misérables*

Les Misérables: each chapter is an hyperevent.

co-appearance of actors in one or more chapters

1	event.id	participant	type
110	01.01.01	MY	chapter
111	01.01.01	NP	chapter
112	01.01.01	MB	chapter
113	01.01.02	MY	chapter
114	01.01.02	ME	chapter
115	01.01.02	MB	chapter
116	01.01.03	MY	chapter
117	01.01.04	MY	chapter
118	01.01.04	ME	chapter
119	01.01.04	CL	chapter
120	01.01.04	GE	chapter
121	01.01.04	MC	chapter
122	01.01.04	MB	chapter
123	01.01.05	MY	chapter
124	01.01.05	MB	chapter
125	01.01.05	ME	chapter



🔗 : Discussion time!

What would be the goal of a RHEM here?

$$E = \{(\{MY, NP, MB\}, t_1), (\{MY, ME, MB\}, t_2), (\{MY\}, t_3), (\{MY, ME, CL, GE, MC, MB\}, t_4), (\{MY, ME, MB\}, t_5), \dots\}$$

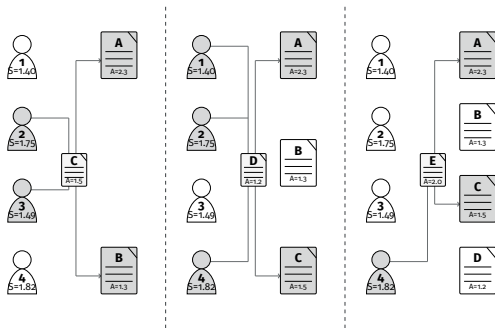
Let us look at each example: 2. Scientific innovation



Scientific innovation: each paper publication is an hyperevent.
a group of authors publishes a paper citing a set of papers

Synthetic data as an example

Time	Authors	Paper	Citations
1950	{2, 3}	<i>C</i>	{ <i>A</i> , <i>B</i> }
1951	{1, 2, 4}	<i>D</i>	{ <i>A</i> , <i>C</i> }
1953	{4}	<i>E</i>	{ <i>A</i> , <i>C</i> }
⋮	⋮	⋮	⋮



🔍 : Discussion time!

Examples of exogenous covariates, here?

$$E = \{(\{2, 3\}, \{A, B\}, t_1 = 1950), (\{1, 2, 4\}, \{A, C\}, t_2 = 1951), (\{4\}, \{A, C\}, t_3 = 1953)\}$$

Relational Hyper Event Modelling (RHEM)



Let $\{N_{SR}(t)\}_{S \subseteq V^S, R \subseteq V^R, t \geq 0}$ be a **counting process** such that:

- ▶ $N_{SR}(t)$ = number of events $S \rightarrow R$ until time t .
- ▶ $N_{SR}(0) = 0$

According to **Doob-Meyer theorem**,

$$N_{SR}(t) = \underbrace{\Lambda_{SR}(t)}_{\text{predictable part}} + \underbrace{M_{SR}(t)}_{\text{noisy part}} = \int_0^t \lambda_{SR}(u) \, du + M_{SR}(t)$$

where $\lambda_{SR}(t)$ is the **instantaneous rate** of $S \rightarrow R$ occurring at time t .

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{SR}(t), t)}_{\text{contribution function}} \right\}$$

⚠ Linear Model: $f(x_{SR}(t), t) = \beta^\top x_{SR}(t) - \beta$ fixed in time and covariate value

Let us focus on $x_{SR}(t)$... from attributes to covariates



$x_{SR}(t)$ should capture information about the instantaneous rate of (S, R, t) .

1. Iterate over specific subsets $S' \subseteq S$.
2. Compute a numerical property $z_{S'}(t)$.
3. Aggregate the values of $z_{S'}(t)$ over the subsets S' .

What does this mean in practice? Let us see two examples!

Exogenous information can be used for several goals...



Node-level feature: $z_s(t)$ is a node attribute, e.g. gender - binary (at the time of Les Misérables...).

Hyperedge covariate: avg. of z_s over nodes in S ; $x_S(t) = \text{avg.z}(S, t) = \sum_{s \in S} z_s(t) / |S|$

Feature (dis)similarity: diff.z measures diversity; a negative parameter $\Rightarrow$ **homophily**.

$$\text{diff.z}(S, t) = \sum_{\{s, s'\} \in \binom{S}{2}} \frac{|z_s(t) - z_{s'}(t)|}{\binom{|S|}{2}}$$



$$\text{avg.z}(S_1) = 0$$

$$\text{diff.z}(S_1) = 0$$



$$\text{avg.z}(S_2) = 0.5$$

$$\text{diff.z}(S_2) = \frac{9}{15}$$



$$\text{avg.z}(S_3) = 1$$

$$\text{diff.z}(S_3) = 0$$

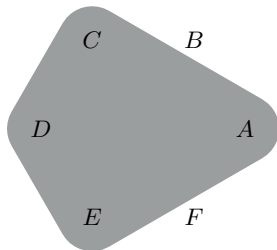


Endogenous $x_{SR}(t)$... Subset-repetition

Subset repetition of order ρ : average # of prior joint events over $S' \subseteq S$ of size ρ .

$$\text{hy.deg}(S', t) = \sum_{e_i: t_i < t} \mathbf{1}(S' \subseteq S_i)$$

$$\text{sub.rep}^{(\rho)}(S, t) = \sum_{S' \in \binom{S}{\rho}} \frac{\text{hy.deg}(S', t)}{\binom{|S|}{\rho}}$$

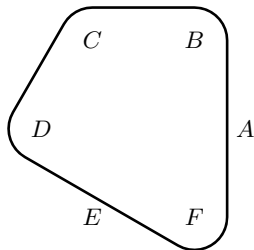


$$S = \{B, C, D, F\}$$

$$\text{sub.rep}^{(1)}(S) = 2/4$$

$$\text{sub.rep}^{(2)}(S) = 1/6$$

$$\text{sub.rep}^{(3)}(S) = 0/4$$



Let's move to $f(x_{SR}(t), t)$... the contribution function



Do you remember?

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{SR}(t), t)}_{\text{contribution function}} \right\}$$

- ▶ We explored **covariates** $x_{SR}(t)$: **structures** based on **past events** or external information.
- ▶ We now focus on their **effect**: their **influence** on the log-event rate.

The simplest form of effect is a **fixed linear effect** (FLE):

$$f^{\text{FLE}}(x_{SR}(t), t) = \beta \cdot x_{SR}(t)$$

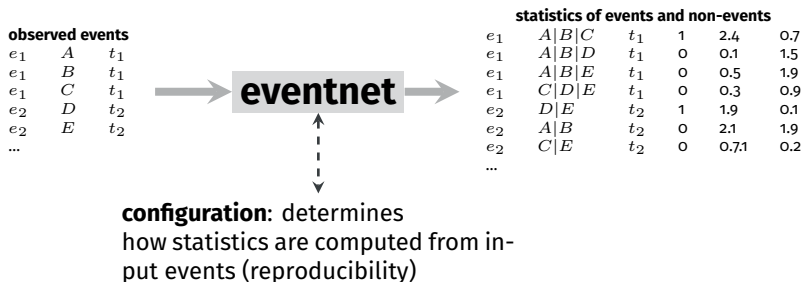
We have already seen that we need to go *beyond linearity*.

What is eventnet?

<https://github.com/juergenlerner/eventnet>



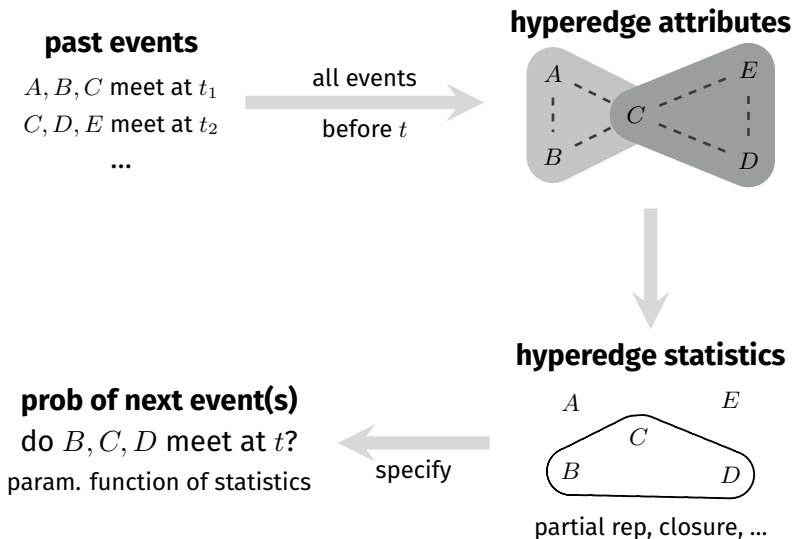
Computes specified covariates for events and non-events.



Configurations can be specified with the eventnet graphical user interface or via a configuration file (XML).



eventnet workflow – more details





Hands-on: Part 1

≈ 45 minutes

- ▶ Format of the input data.
- ▶ CSV and time settings.
- ▶ Specifying network attributes and statistics.
- ▶ Specifying the observed events, risk set, and sampling.
- ▶ Perform case-control partial likelihood inference.
- ▶ Fit a linear RHEM in R.

[https://martinaboschi.github.io/rem-tutorial/
02-All-you-need-to-know-RHEM/01-Les-Miserables/](https://martinaboschi.github.io/rem-tutorial/02-All-you-need-to-know-RHEM/01-Les-Miserables/)

Directed hyperevents



Research settings about

- ▶ **teams of actors** collaboratively doing **work**
- ▶ building on **prior work** (of the same or different actors).

Examples include:

- ▶ Scientists publishing papers that cite previous papers.
- ▶ Inventors filling patents referencing other patents.
- ▶ Artists producing music / movies making implicit or explicit references to other music / movies.

Case study



Data: <https://www.aminer.org/citation>.

DBLP-Citation-network V14, restricted to journal papers.

1,416,353 papers written by 1,286,941 unique authors

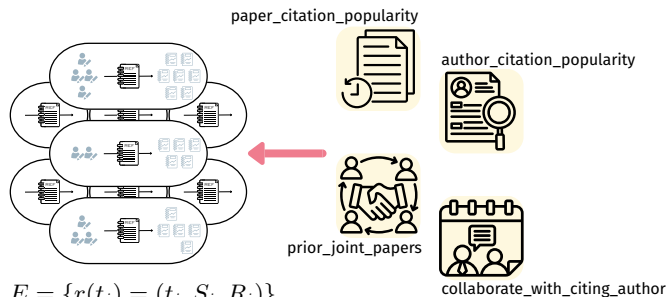
References:

1. Lerner, Hâncean, & Lomi: **Relational hyperevent models for the coevolution of coauthoring and citation networks.**
[Lerner et al., 2025]
2. Boschi, Lerner, & Wit: **Beyond linearity and time-homogeneity: Relational hyper event models with time-varying non-linear effects.**
[Boschi et al., 2026]

Drivers of Scientific Innovation

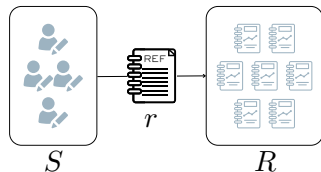


Relational Hyperevent: publication in year t of article $r(t)$ by a group of authors S citing articles R



$$E = \{r(t_i) = (t_i, S_i, R_i)\}$$

$$i = 1, \dots, 1\,416\,353$$

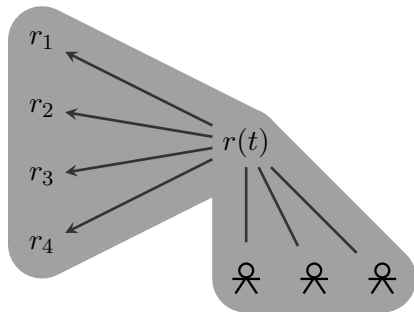


Relational Hyper Event Model: captures how key covariates **influence** the rate of hyperevent occurrences.



The unit of analysis.

Work $r(t)$ produced at time t by **team** of actors $S = \{s_1, s_2, s_3, \dots\}$ referencing a **list** of previous works $R = \{r_1, r_2, r_3, r_4, \dots\}$.



Previous works produced by same, overlapping or disjoint teams; referencing other previous works, produced by ...



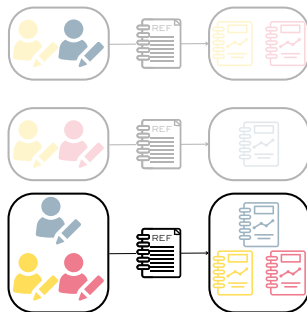
Subset repetition of order (ρ, ℓ)

Covariates $x_{SR}(t)$: statistics derived from **past events** and/or **exogenous information**.
How do these covariates change in the case of **directed hyperevents**?

► : E.g. **Subset repetition**:

$$\text{cite}^{\text{aut-pap}}(S, R, t) = \sum_{t_i < t} \omega(t - t_i) \cdot 1_{\{S \subseteq S_i \cap R \subseteq R_i\}}$$

$$\text{subrep}^{(\rho, \ell)}(S, R, t) = \sum_{(S', R') \in \binom{S}{\rho} \times \binom{R}{\ell}} \frac{\text{cite}^{\text{aut-pap}}(S', R', t)}{\binom{|S|}{\rho} \times \binom{|R|}{\ell}}$$





Types of effects / research questions.

Effects in the collaboration network.

- ▶ Activity of authors; homophily effects.
- ▶ Repeated collaboration among pairs, triples, ...of authors.
- ▶ Closure: collaborating with collaborators' collaborators.

Effects in the reference network.

- ▶ Popularity of papers.
- ▶ Repeated co-references to pairs, triples, ...of papers.

Effects in the mixed two-mode network.

- ▶ Referencing the own paper (self-citation).
- ▶ Referencing the papers/references of collaborators.
- ▶ Collaborating with authors referencing own/same papers.
- ▶ Reference repetition / reciprocation on the author level.

E.g. **cite paper and its references** (whiteboard).



Before the practice...

Do you remember?

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{SR}(t), t)}_{\text{contribution function}} \right\}$$

- ▶ We explored **covariates** $x_{SR}(t)$: **structures** based on **past events** or external information.
- ▶ We now focus on their **effect**: their **influence** on the log-event rate.

The simplest form of effect is a **fixed linear effect** (FLE):

$$f^{\text{FLE}}(x_{SR}(t), t) = \beta \cdot x_{SR}(t)$$

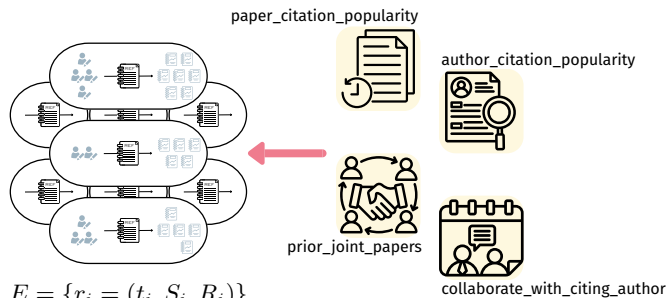
We have already seen that we need to go **beyond linearity**.

	parameters
num prior papers	0.129 (0.003)***
diff prior papers	0.185 (0.002)***
num prior joint papers	0.823 (0.004)***
author citation popularity	−0.017 (0.001)***
diff auth cite pop	−0.007 (0.001)***
collab w citing author	0.318 (0.003)***
closure: common coauthor	−0.014 (0.001)***
closure: citing common paper	−0.112 (0.002)***
closure: citing common author	0.001 (0.001)
closure: cited by common author	0.007 (0.001)***
paper citation popularity	0.170 (0.000)***
paper-pair cocitation	0.032 (0.001)***
paper-triple cocitation	−0.170 (0.004)***
author citation repetition	0.643 (0.008)***
paper outdegree popularity	0.195 (0.001)***
cite paper and its refs	7.360 (0.010)***
adopt citation of coauthor	−0.264 (0.003)***
self citation	1.799 (0.012)***
cite coauthor's paper	0.291 (0.007)***
author-author citation repetition	−0.381 (0.007)***
author-author citation reciprocation	−0.343 (0.007)***

Are drivers of Scientific Collaboration truly Constant?

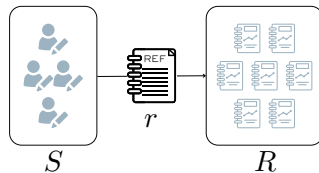


Relational Hyperevent: publication in year t of article $r(t)$ by a group of authors S citing articles R



$$E = \{r_i = (t_i, S_i, R_i)\}$$

$$i = 1, \dots, 1\,416\,353$$

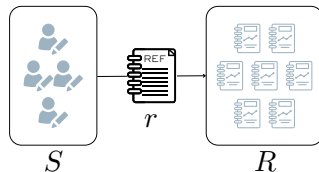
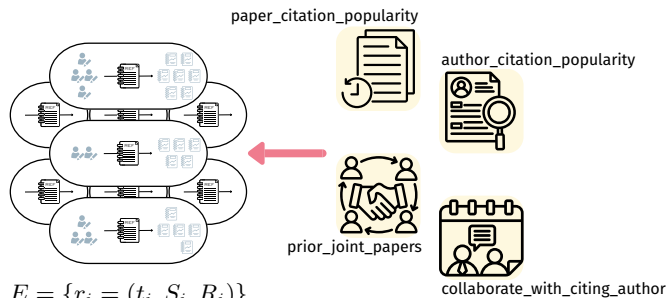


Relational Hyper Event Model: captures how key covariates **influence** the rate of hyperevent occurrences.

Are drivers of Scientific Collaboration truly Constant?



Relational Hyperevent: publication in year t of article $r(t)$ by a group of authors S citing articles R



Relational Hyper Event Model: captures how key covariates **influence** the rate of hyperevent occurrences.

🔍 : Discussion time! Any idea why this should not be the case?

What can break Linearity and Time-Homogeneity?

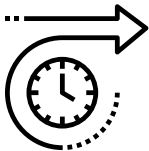


Saturation: individuals face cognitive overload and become **unable to effectively additional information**.

💡: *allow drivers' influence to exhibit a non-linear pattern.*

Forgetfulness: propensity to **lose track of past details** over time.

💡: *define mechanisms as interval times with non-linear effect.*



External Changes: **shifts** in the environment, technology, or social norms that **alter how factors influence system**.

💡: *allow drivers' influence to have a time-varying effect.*



Back to the hazard

$$\lambda_{SR}(t) = \underbrace{W_{SR}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{SR}(t), t)}_{\text{contribution function}} \right\}$$

$$f(x_{sr}(t), t) = \underbrace{\delta \cdot x_{sr}^{(1)}(t)}_{\text{fixed linear}} + \underbrace{f(x_{sr}^{(2)}(t))}_{\text{non-linear}} + \underbrace{\alpha(t) \cdot x_{sr}^{(3)}(t)}_{\text{time-varying}}$$

1. **fixed linear**: δ is time-homogenous and is not a function of $x_{SR}^{(1)}(t)$;
2. **non-linear**: $f(x_{SR}^{(2)}(t))$ is a non-linear function of $x_{SR}^{(2)}(t)$;
3. **time-varying**: $\alpha(t)$ is a non-linear function of time;

Modeling Non-Linear Effects



- ▶ Effect of $x_{SR}^{(2)}(t)$ as $\rightarrow$ **Spline function of Covariate**

$$f(x_{SR}^{(2)}(t)) \approx \sum_{j=1}^q \theta_j b_j(x_{SR}^{(2)}(t))$$

Choices to be made:

- ▶ **basis dimension** q
- ▶ a **set of smooth basis functions** $\{b_j\}_j$ e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate q **smooth non-linear functions** of covariate $b_j(x_{SR}^{(2)}(t))$.

Estimating $f \iff$ Estimating β_j -s



How NLE Enters the Model and Likelihood

► ... to **log-hazard of an interaction**

$$f(\mathbf{x}_{SR}(t), t) = \dots + \underbrace{\sum_{j=1}^q \beta_j b_j(x_{SR}^{(2)}(t))}_{\text{non-linear effect}} + \dots$$

► ... to **sampled likelihood function (m=1)**

$$\mathcal{L}^S(\boldsymbol{\theta}) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$



Contribution of a covariate with NLE

► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$

stratum	time	sender_group_ev	receiver_group_ev	sender_group_nv	receiver_group_nv	avg_team_vuln_ev	avg_team_vuln_nv
1	0.02	{S2,S3}	{R2}	{S1,S2,S4}	{R3,R4}	4.01	1.88
2	0.07	{S3,S4}	{R1,R3,R4}	{S1,S4}	{R1,R2,R4}	2.54	3.26
3	0.15	{S2,S3}	{R2,R4}	{S2,S3,S4}	{R2,R4}	2.96	2.96
4	0.76	{S1,S2,S3,S4}	{R2,R4}	{S1,S3,S4}	{R3,R4}	2.96	1.88
5	0.76	{S3}	{R1,R3,R4}	{S2,S3,S4}	{R1,R3,R4}	2.54	2.54
6	0.76	{S3}	{R1,R3,R4}	{S1,S2}	{R3,R4}	2.54	1.88

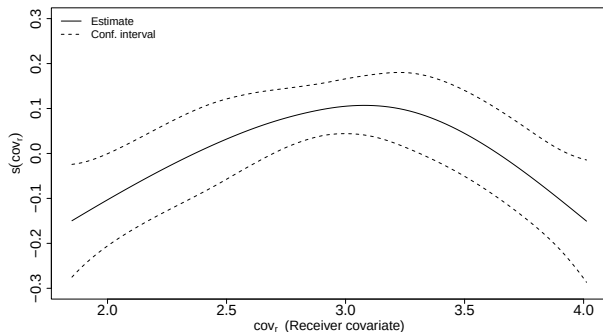
```
rem(~ nl(avg_team_vuln), method = "gam", data = data_nle_r)
```



Contribution of a covariate with NLE

► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \beta_j [b_j(x_{S_i R_i}^{(2)}(t_i)) - b_j(x_{S_i^* R_i^*}^{(2)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$





Modeling Time-Varying Effects

- ▶ Effect of $x_{SR}^{(3)}(t)$ as $\rightarrow$ **Spline function of Time**

$$\alpha(t) \approx \sum_{j=1}^q \gamma_j a_j(t)$$

Choices to be made:

- ▶ **basis dimension** q
- ▶ a **smooth basis functions** $\{a_j\}_j$ e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate q **smooth non-linear functions** of time $a_j(t)$.

Estimating $\alpha(t) \iff$ Estimating γ_j -s



How TVE Enters the Model and Likelihood

► ... to log-hazard of an interaction

$$f(\mathbf{x}_{SR}(t), t) = \dots + \underbrace{\sum_{j=1}^q \gamma_j a_j(t)}_{\text{time-varying effect}} \times \underbrace{x_{SR}^{(3)}(t)}_{\text{covariate}} + \dots$$

► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \alpha(t_i)[x_{S_i R_i}^{(3)}(t_i) - x_{S_i^* R_i^*}^{(3)}(t_i)] + \dots}}{1 + e^{\dots + \alpha(t_i)[x_{S_i R_i}^{(3)}(t_i) - x_{S_i^* R_i^*}^{(3)}(t_i)] + \dots}} \quad \downarrow$$

$$\mathcal{L}^S(\gamma) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}} \quad (+ \text{ smoothing penalty})$$



Contribution of a covariate with TVE

► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\gamma) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}} \quad (+ \text{ smoothing penalty})$$

stratum	time	sender_group_ev	receiver_group_ev	sender_group_nv	receiver_group_nv	avg_team_aggres_ev	avg_team_aggres_nv
1	0.02	{S2,S3}	{R2}	{S1,S2,S4}	{R3,R4}	6.02	3.36
2	0.07	{S3,S4}	{R1,R3,R4}	{S1,S4}	{R1,R2,R4}	4.05	2.34
3	0.15	{S2,S3}	{R2,R4}	{S2,S3,S4}	{R2,R4}	6.02	4.51
4	0.76	{S1,S2,S3,S4}	{R2,R4}	{S1,S3,S4}	{R3,R4}	4.18	3.77
5	0.76	{S3}	{R1,R3,R4}	{S2,S3,S4}	{R1,R3,R4}	6.63	4.51
6	0.76	{S3}	{R1,R3,R4}	{S1,S2}	{R3,R4}	6.63	4.30

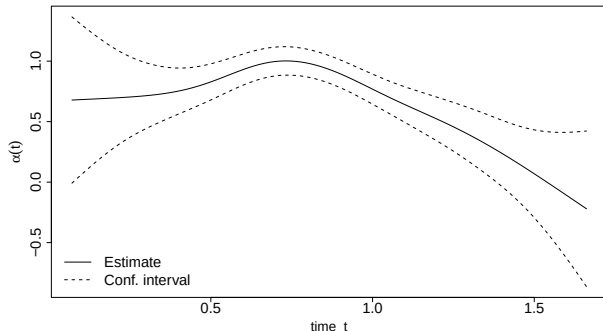
```
rem(~ tv(avg_team_aggres), time = "time", method = "gam", data = data_tve_s)
```



Contribution of a covariate with TVE

► ... to **sampled partial likelihood (m=1)**

$$\mathcal{L}^S(\gamma) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \gamma_j a_j(t_i) \right] \Delta x_i^{(3)} + \dots}} \quad (+ \text{ smoothing penalty})$$





TVE and NLE for RHEMs

Contribution function: impact of the covariate on the log-rate of event occurrence.

$$f(x_{SR}(t), t)$$

- Fixed linear effect (FLE):

$$f^{\text{FLE}}(x_{SR}(t), t) = \delta \cdot x_{SR}(t)$$

- Non-linear effect (NLE): captures *saturation* and *forgetfulness*.

$$f^{\text{NLE}}(x_{SR}(t)) = \sum_{q=1}^Q \beta_q \underbrace{b_q(x_{SR}(t))}_{\text{basis function evaluation of covariate}}$$

- Time-varying effect (TVE): an *effect may change over calendar time* due to *changing circumstances* or *unmeasured confounders*.

$$f^{\text{TVE}}(x_{SR}(t), t) = \alpha(t) \cdot x_{SR}(t), \quad \alpha(t) = \sum_{l=1}^L \gamma_l \underbrace{a_l(t)}_{\text{basis function evaluation of time}}$$

Joint time-varying non-linear effect...



💡 Proposal: employ **tensor product smooth** (Wood, 2016) to model TVNLE in RHEM.

$$f^{\text{TVNLE}}(x_{SR}(t), t) = \sum_{q=1}^Q \left(\underbrace{\sum_{l=1}^L \gamma_{ql} a_l(t)}_{\text{time-varying coefficient}} \right) \underbrace{b_q(x_{SR}(t))}_{\text{basis function evaluation of covariate}}$$

😊: coefficients of basis evaluations of covariates are not fixed but vary over time.

Joint time-varying non-linear effect...



💡 Proposal: employ **tensor product smooth** (Wood, 2016) to model TVNLE in RHEM.

$$f^{\text{TVNLE}}(x_{SR}(t), t) = \sum_{q=1}^Q \left(\underbrace{\sum_{l=1}^L \gamma_{ql} a_l(t)}_{\text{time-varying coefficient}} \right) \underbrace{b_q(x_{SR}(t))}_{\text{basis function evaluation of covariate}}$$

😊: coefficients of basis evaluations of covariates are not fixed but vary over time.

❓ : **Discussion time!** How would you represent this type of effect?

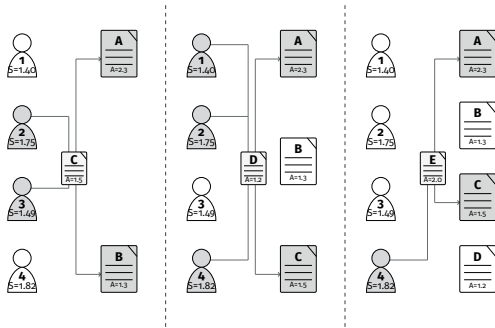
Joint time-varying non-linear effect... in practice I



Scientific innovation: each paper publication is an hyperevent.
a group of authors publishes a paper citing a set of papers

Synthetic data as an example

Time	Authors	Paper	Citations
1950	{2, 3}	<i>C</i>	{ <i>A</i> , <i>B</i> }
1951	{1, 2, 4}	<i>D</i>	{ <i>A</i> , <i>C</i> }
1953	{4}	<i>E</i>	{ <i>A</i> , <i>C</i> }
⋮	⋮	⋮	⋮

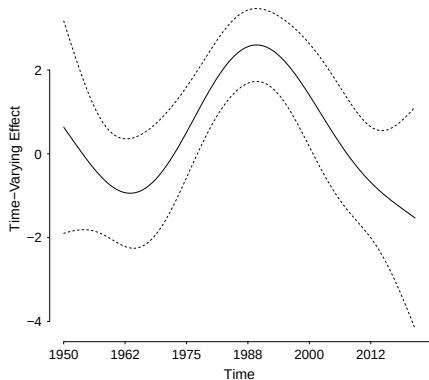


$$E = \{(\{2, 3\}, \{A, B\}, t_1 = 1950), (\{1, 2, 4\}, \{A, C\}, t_2 = 1951), (\{4\}, \{A, C\}, t_3 = 1953)\}$$

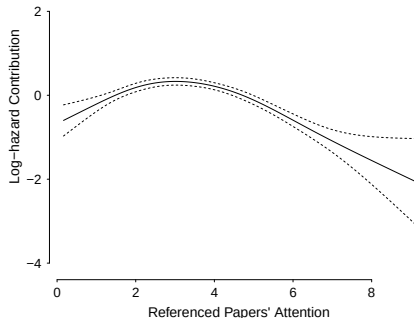


Let's start with TVE and NLE...

Time-Varying Effect of Average Author Score



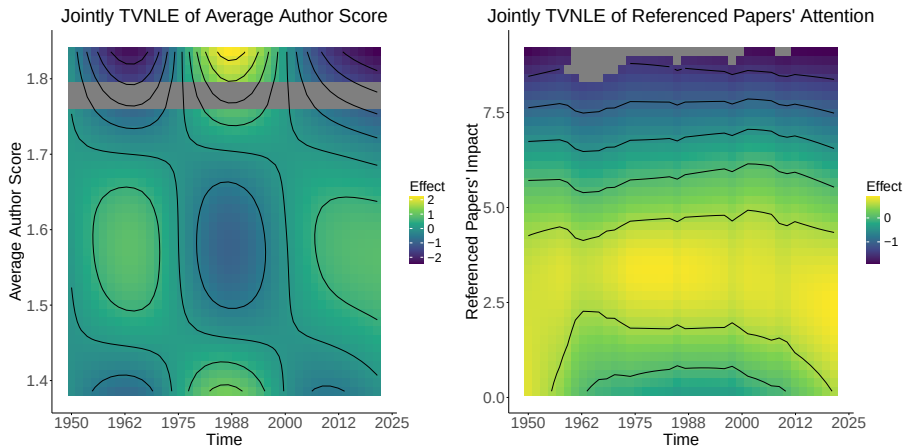
Non-linear Effect of Referenced Papers' Attention



🔗 : Discussion time! Do you remember how to fit these effects in R Studio?
How do we interpret them?



Let's now fit a TVNLE...



🔍 : **Discussion time!** Do we need a TVNLE effect in both cases?



Step 1: compute covariates with `eventnet`

Using the `eventnet` software:

1. Extract event information from the raw data.
2. Specify the components required by `eventnet` to define the risk set.
3. Sample a non-event observation from the risk set.
4. Compute covariates for both events and non-events.

Case-control dataset: Merge the output of `eventnet` by event and non-event identifiers so that each row contains information for both event and non-event.



Step 2: compute covariates with `eventnet`

1. Apply a monotonic transformation to each covariate x :

$$\dot{x} = 1 - \exp\left(-\frac{x}{\epsilon}\right)$$

- Choose ϵ so that the transformed covariate (evaluated for events) is as close as possible to a uniform distribution.
- Apply the same transformation to both events and non-events.

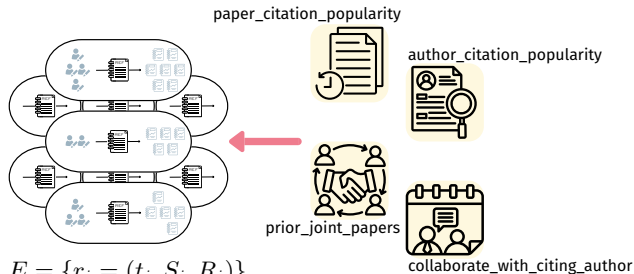
2. Transform event times t using the empirical cumulative distribution function:

$$\dot{t} = \text{ecdf}_t(t)$$

Coauthorship Citation Networks



RHEM extension applied to the DBLP-Citation-network V14 dataset (journal articles).



$$E = \{r_i = (t_i, S_i, R_i)\}$$

$$i = 1, \dots, 1\,416\,353$$

1. Prior papers ;
2. Prior joint papers;
3. Paper citation popularity ;
4. Paper pair co-citation;
5. Author citation repetition;
6. Cite paper and its references;
7. Difference in prior papers ;
8. Author citation popularity;
9. Difference in author citation popularity;
10. Collaborate with citing author;
11. Author self citation ;
12. Paper outdegree popularity ;



Author Self Citation - Covariate

► STEP 1: Define the k -th covariate.



$$\text{author}(s, r, t) = \sum_{t_i < t} \omega(t - t_m) \cdot \mathbb{1}_{\{s=j_i \wedge r \in S_i\}}$$

$$x^{(k)} = \text{author_self_citation}(S, R, t) = \sum_{s \in S, r \in R} \frac{\text{author}(s, r, t)}{|S| \cdot |R|}$$

► STEP 2:

- **Transform the k -th covariate**, to *improve model fitting stability*.

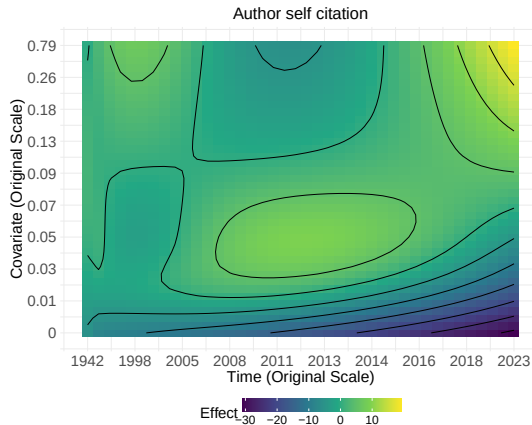
$$\dot{x}^{(k)} = 1 - \exp\left(-\frac{x^{(k)}}{\epsilon^{(k)}}\right),$$

where $\epsilon^{(k)}$ is selected through an optimisation procedure.

- **Transform time**, to *improve model fitting stability*.

$$\dot{t} = \text{ecdf}_t(t)$$

Author Self Citation - Effect



- ▶ monotone effect at the beginning and end of the observation period, only.
- ▶ **between 2008 and 2016, the effect is non-monotonic:**
it first increases with the covariate value, then decreases.



Hands-on: part 2

≈ 40 minutes

- ▶ Load the data in R Studio.
- ▶ Create the case-control dataset.
- ▶ Fit models with NLE, TVE, and TVNLE.

[https://martinaboschi.github.io/rem-tutorial/
02-All-you-need-to-know-RHEM/02-Scientific-Innovation/](https://martinaboschi.github.io/rem-tutorial/02-All-you-need-to-know-RHEM/02-Scientific-Innovation/)

Today was quite an intense RHEM day...



- ▶ Tensor product smooths can be employed to **model TVNLE dynamics in RHEMs**.
- ▶ Joint TVNLE models are a powerful tool for **uncovering not only non-linear but also non-monotonic patterns** that **linear models would fail to capture**.
- ▶ Applied this approach to study the **dynamics of citation and impact**.
- ▶ **Author self-citation:**
 - Citing a certain share of one's own work seems natural and nearly unavoidable;
 - Excessive self-citation could be perceived negatively by the research community.



Relational events that have an outcome.

Sometimes, relational (hyper)events produce a tangible outcome with associated “success” or “performance”.

- ▶ Teams of scientists publishing an article → impact.
- ▶ Filmmakers producing a film → revenue, popularity, evaluation, awards.
- ▶ Medical teams performing an operation → cost, duration, benefits for the patient.
- ▶ Two soccer teams meeting in a match.

Past success may explain future collaboration (RHEM).

Past success may explain future success (RHOM).

Lerner & Hâncean (2023). **Micro-level network dynamics of scientific collaboration and impact: relational hyperevent models for the analysis of coauthor networks.** *Network Science*.

[[Lerner and Hâncean, 2023](#)]

Data: relational events that have an outcome.



A relational hyperevent with an outcome is given by

$$e_i = (S_i, t_i, x_i, y_i) \text{ ,}$$

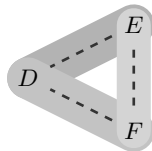
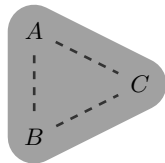
- ▶ t_i is the time of the event;
- ▶ $S_i \subseteq V_{1,t_i}$ event participants (e. g., coauthors);
- ▶ x_i covariates of the event (type, weight);
- ▶ y_i is the *relational outcome* (success, performance, impact).



Recall: subset repetition of order k .

Average number of prior joint events over $S' \subseteq S$ of size S .

$$\text{sub.rep}^{(k)}(S, t) = \sum_{S' \in \binom{S}{k}} \frac{\text{hy.deg}(S', t)}{\binom{|S|}{k}}.$$



$$\text{sub.rep}^{(1)}(\{A, B, C\}) = 1$$

$$\text{sub.rep}^{(2)}(\{A, B, C\}) = 1$$

$$\text{sub.rep}^{(3)}(\{A, B, C\}) = 1$$

$$\text{sub.rep}^{(1)}(\{D, E, F\}) = 2$$

$$\text{sub.rep}^{(2)}(\{D, E, F\}) = 1$$

$$\text{sub.rep}^{(3)}(\{D, E, F\}) = 0$$

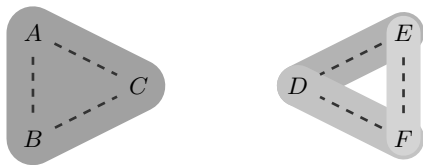
$$\text{hy.deg}(S', t) = \sum_{e_i: t_i < t} \mathbf{1}(S' \subseteq S_e)$$



Prior (shared) success of order k .

Average weight of prior joint events over $S' \subseteq S$ of size k .

$$\text{prior.success}^{(k)}(S, t) = \frac{\sum_{S' \in \binom{S}{k}} \text{performance}(S', t)}{\sum_{S' \in \binom{S}{k}} \text{hy.deg}(S', t)} .$$



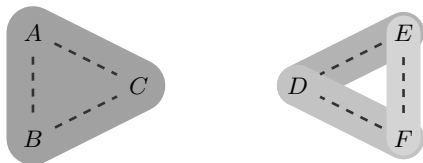
$$\text{performance}(S', t) = \sum_{e_i: t_i < t} y_i \cdot \mathbf{1}(S' \subseteq S_i) .$$

Related: prior success disparity.



Individual success vs. shared success.

Assume that two teams are composed of individuals that have been equally successful in the past.



It still makes a difference whether they have **jointly** achieved this success.

A team is more than the sum of its members.

Recall: relational events that have an outcome.



A relational hyperevent with an outcome is given by

$$e_i = (S_i, t_i, y_i) ,$$

- ▶ t_i is the time of the event;
- ▶ $S_i \subseteq V_{1,t_i}$ event participants (e. g., coauthors);
- ▶ y_i is the *relational outcome* (success, performance, impact).

RHEM explain the participants S_i (and/or the time t_i).

RHOM explain the outcome y_i , given that there is an event (S_i, t_i) .

Relational hyperevent outcome models (RHOM).



Given of sequence of relational hyperevents with an outcome

$$E = [(S_1, t_1, y_1), \dots, (S_n, t_n, y_n)]$$

RHOM specify regression models (OLS, logit, ...) for

$$y_i \sim f(t_i, S_i; \mathcal{W}_{t_i^-})$$

Allows to control for certain forms of dependence:

- ▶ repeated observations by the same team;
- ▶ observations with overlapping teams;

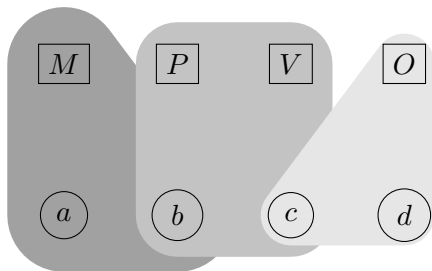
R(H)EM and R(H)OM can be specified with the same explanatory variables, allowing to compare effects on **team selection** and **team performance**.



Relational events that are associated with labels.

Actors co-participating in crime events of certain type:
market (M), property (P), violence (V), “other” (O).

$$\begin{aligned}e_1 &= ([\{a, b\}, \{M\}], t_1) \\e_2 &= ([\{b, c\}, \{P, V\}], t_2) \\e_3 &= ([\{c, d\}, \{O\}], t_3)\end{aligned}$$



In general, meeting, communication, or team work events might have associated “labels” (purposes, objectives, topics, ...).

Two-mode relational hyperevent network.

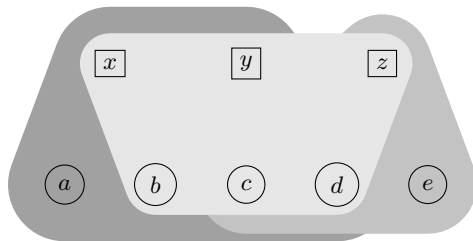


Actors (circles: $a, b, \dots$) participate in events having labels (squares: $x, y, \dots$).

$$e_1 = ([\{a, b, c, d\}, \{x, y\}], t_1)$$

$$e_2 = ([\{c, d, e\}, \{z\}], t_2)$$

$$e_3 = ([\{b, c, d\}, \{x, y, z\}], t_3)$$



Many additional effects possible:

Partial repetition: of actors, labels, and mixed.

Closure: in the space of actors, labels, and mixed.



References I

- Martina Boschi, Jürgen Lerner, and Ernst C. Wit. Beyond linearity and time-homogeneity: Relational hyper event models with time-varying non-linear effects. *Social Networks*, 86:120–137, 2026. ISSN 0378-8733. doi: <https://doi.org/10.1016/j.socnet.2026.02.001>.
- Jürgen Lerner and Marian-Gabriel Hâncean. Micro-level network dynamics of scientific collaboration and impact: Relational hyperevent models for the analysis of coauthor networks. *Network Science*, 11(1):5–35, 2023.
- Jürgen Lerner, Marian-Gabriel Hâncean, and Alessandro Lomi. Relational hyperevent models for the coevolution of coauthoring and citation networks. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 188(2):583–607, 2025.